# Formula 1 Podium Prediction

## Milestone 1 — Analytical Report

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Data: Kaggle / Ergast Formula 1 World Championship dataset (1950–2024, 14 tables) · Notebook: “F1 Podium Prediction – Milestone 1 – Team 11” (Kaggle, version 8, runs top to bottom in about 6 minutes)

**In one paragraph.** For every driver in every race we predict whether he finishes in the top 3, using only what is known after qualifying and before the start. We cleaned the 14 linked tables (sentinels, types, keys, one row per driver and race), answered the three data-engineering questions, built 20 leakage-safe features in 6 groups for the seasons 2003–2024, and split the data by season (train 2003–2018, validation 2019–2021, test 2022–2024). Of 13 modelling attempts, a tuned random forest was best on validation (PR-AUC 0.748) and reached **PR-AUC 0.719 on the unseen 2022–2024 seasons**, against 0.647 for the simple rule “podium if grid ≤ 3”. The starting grid carries most of the signal; recent form, standings and qualifying pace add to it. All explanations describe what the model relies on, **not** what causes podiums.

## 1. Task, target and prediction cutoff

- **Unit of analysis:** one driver entering one race; the pair (raceId, driverId) is unique in every modelling row.
- **Target:** `podium = positionOrder ≤ 3` (binary). `positionOrder` is never missing, so every row has a label.
- **Cutoff:** after qualifying, before the race starts. Results, points, status, laps, pit stops, lap times and standings *after* the race are never features. Standings tables for race X already include race X, so only the previous round is used.
- **Class balance:** 13.6% of starters finish on the podium (14.4% from 2003). A model that always says “no” would be about 86% accurate and useless — this is why **PR-AUC is the main metric** and ROC-AUC the secondary one (Decision 3); F1, precision and recall are reported at a threshold chosen on validation.

## 2. Data audit and cleaning

**Loading (Decision 13).** All 14 CSVs are read as-is, so the sentinel `\N` stays visible as text in `raw` (160,076 cells in 6 tables) together with 68 empty cells in `qualifying.q2/q3`. Cleaning turns all 160,144 markers into real missing values (NaN) on copies; `raw` is never changed (checked by a fingerprint at the end of cleaning and again after Section 9).

**Semantic typing (Decision 14).** 24 columns are converted in place: 10 to numbers, 7 to dates, and 7 lap or pit times to seconds with a helper `time_to_seconds()`, which equals `milliseconds / 1000` on all 589,081 laps and 11,371 pit stops.

**Keys.** 41 checks (14 primary keys, 4 natural keys, 23 foreign keys): 0 empty keys, 0 duplicates, 0 broken links.

**Rows removed (applied in this order):**

| Step | Rule and reason | results rows |
|---|---|---|
| start | raw `results` | 26,759 |
| Decision 6 | Drop the 11 Indianapolis 500 races (1950–1960): a separate series — only 0.6% of its drivers raced another GP that season, 99.8% American (would inflate Q2) | 26,354 |
| Decision 8 | Drop drivers who never started (`positionText` F = failed to qualify, W = withdrew): 1,704 rows. Keep the 87 pit-lane starters (grid = 0) | 24,650 |
| Decision 5 | Shared drives (1950s): drop every row of a repeated (raceId, driverId) pair. The grid cannot tell the cars apart, and “keep the first row” would choose by finishing order (resultId follows it in 1,074 of 1,125 races) — leakage | **24,525** |

Cleaning deliberately does **not** fill missing values or drop rows with gaps: filling belongs in the model Pipeline, fitted on training data only, and `dropna()` would remove every driver who did not finish.

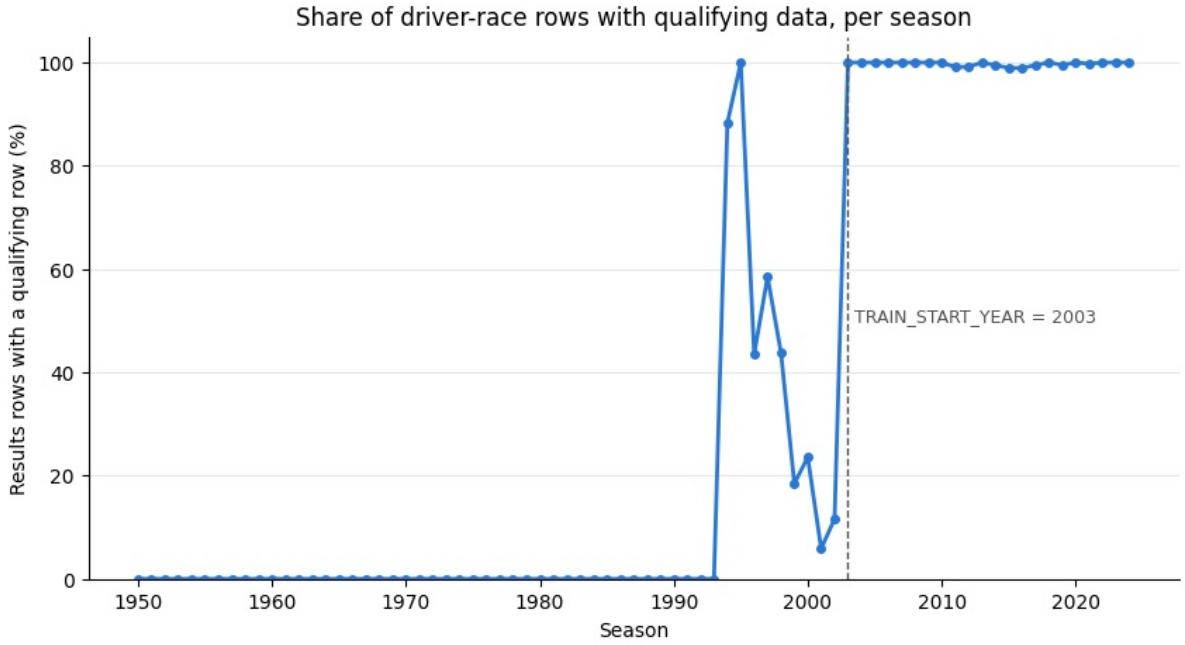



**Figure 1.** Qualifying coverage per season: none before 1994, patchy 1996–2002 (down to 6% in 2001), at least 98.9% from 2003. This is why the modelling table starts in 2003 (Decision 4), later confirmed by an experiment (Section 8).

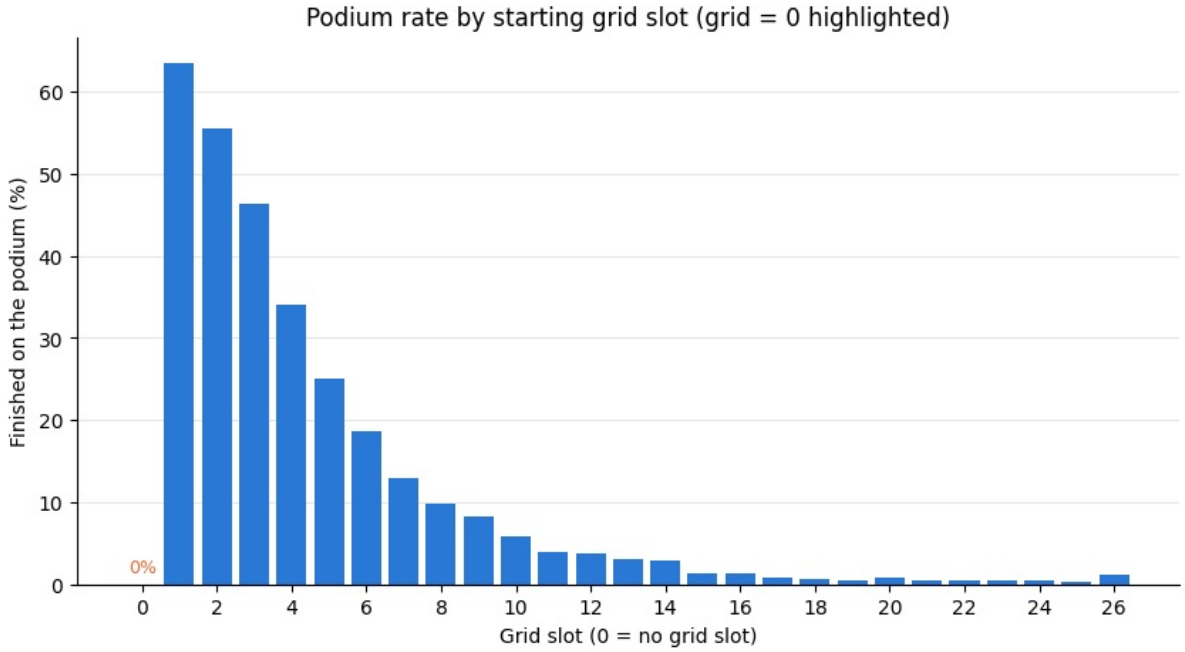



**Figure 2.** Podium rate by grid slot (raw data): 63.5% from pole, under 6% from 10th, 0% for grid = 0 — the strongest single signal and the evidence for the grid features and the “grid ≤ 3” rule baseline.

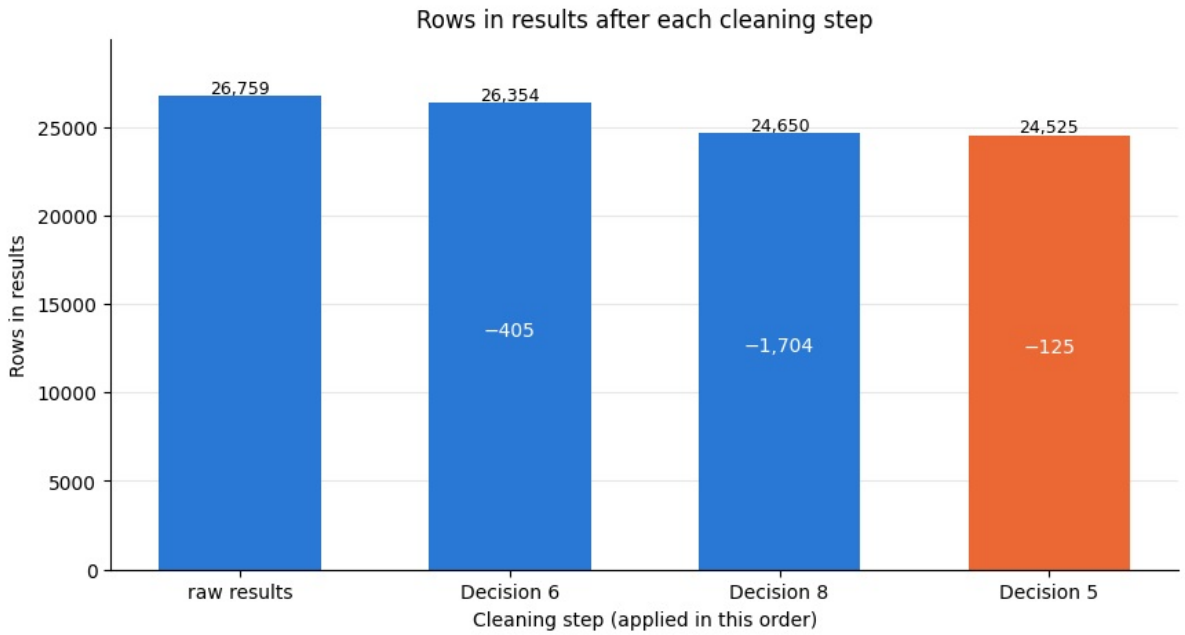



**Figure 3.** EDA after cleaning: rows per table before and after cleaning. Five of the 14 tables lose rows (only Decisions 6, 8 and 5 remove rows).

**EDA before vs after.** The `grid = 0` spike shrinks from 1,638 rows (6.1%) to the 87 pit-lane starters; entries exceeded starters by up to 12.7 drivers per race (1989) and after cleaning every row is a starter; the podium share of rows rises from 12.7% to 13.6%. Retirement rates fell from 40–50% of starters before 2000 to about 12% in the 2020s — eras differ strongly, which supports starting in 2003 and splitting by time.

## 3. Data-engineering questions

All three use one joined table (`analysis`: results → races → circuits, drivers, constructors, status; 24,525 rows, grain asserted, every join many-to-one).

### Q1. Which circuits convert a front-row start into a podium most often, and does the ranking change after 2014?

**Definition:** front row = **grid** slot 1 or 2 (per the TA: the question is about the starting position, not qualifying); rate = podiums ÷ front-row starts, per circuit, before 2014 vs from 2014. **Grid vs qualifying:** they differ in 23.3% of driver-races since 1994 (1.5% in 1994–2003, 25.4% in 2004–2013, 30.4% from 2014, because penalties move drivers back); 3.8% of drivers who qualified on the front row did not start there. **Sample rule (Decision 15):** only circuits with at least 10 front-row starts in each period are ranked (16 circuits), with 95% Wilson intervals.

**Answer.** Over all circuits, front-row conversion rose from **56.7%** before 2014 to **76.2%** from 2014. Highest from 2014: Monaco 89.5%, Suzuka 88.9%, Abu Dhabi 86.4%; lowest: Budapest 59.1%, Mexico City 61.1%, Sakhir 66.7%. The ranking changes a lot (Spearman −0.08, bootstrap interval −0.50 to 0.38; 0.36 with 2003–2013 as the “before” period) — Monaco climbs from 15th to 1st, Budapest falls from 2nd to 16th — but with only 10–26 starts per circuit after 2014, **only Monaco’s change is clearly larger than chance**. *Grain:* driver-race. *Missingness:* grid is complete. *Cutoff:* the grid is known before the start.

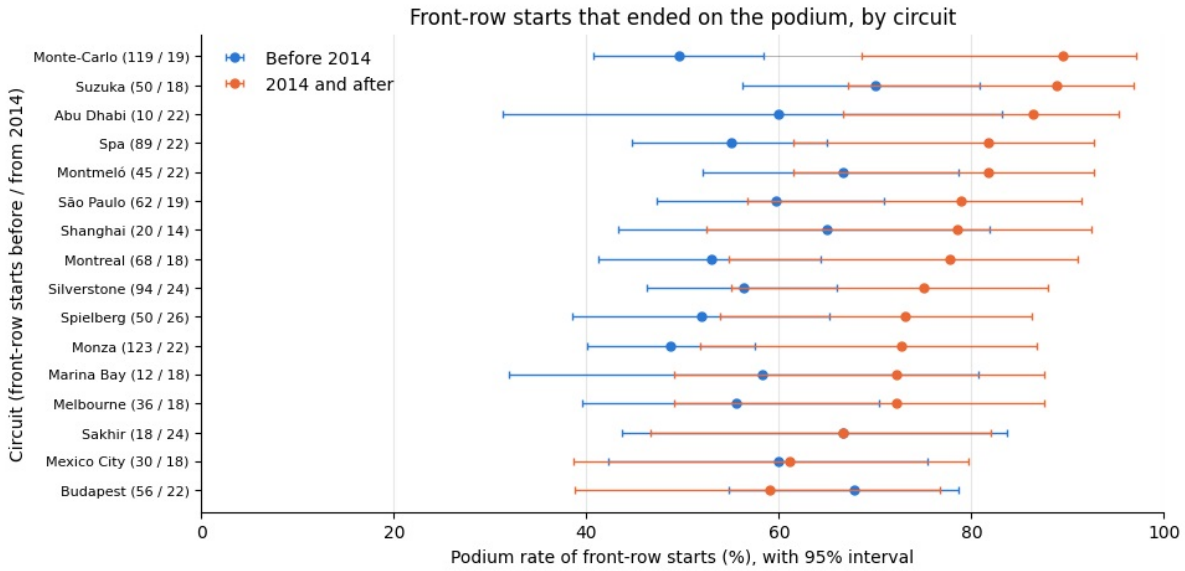



**Figure 4.** Q1: front-row → podium rate per circuit, before 2014 (blue) vs from 2014 (orange), with 95% Wilson intervals.

### Q2. Do drivers podium more often at home, after controlling for starting position?

**Home flag (Decision 12):** an explicit nationality → country dictionary (43 nationalities, all mapped and asserted; dual nationalities are home in either country; one raw value "Argentinian " had a trailing space; Las Vegas is spelled "United States"). Home race in 1,914 of 24,525 rows (7.8%). **Design (Decision 16):** compare home and away drivers *inside* four grid buckets (1–3, 4–6, 7–10, 11+), so drivers who started from similar places are compared; pit-lane starts are left out; one pooled difference re-weights away drivers to the home drivers' grid mix; the interval comes from a bootstrap over whole races.

**Answer.** Inside the buckets the rates are nearly identical (grid 1–3: 57.1% home vs 55.8% away; 4–6: 26.9% vs 26.1%; 7–10: 7.2% vs 9.3%; 11+: 1.1% vs 1.7%). The adjusted difference is **−0.5 points (95% interval −1.7 to +0.8)**; for 2003–2024 −1.4 (−4.1 to +1.5). **There is no sign of a home advantage once the starting position is controlled** (an association, not a causal claim). *Cutoff:* nationality, circuit and grid are all known before the race.

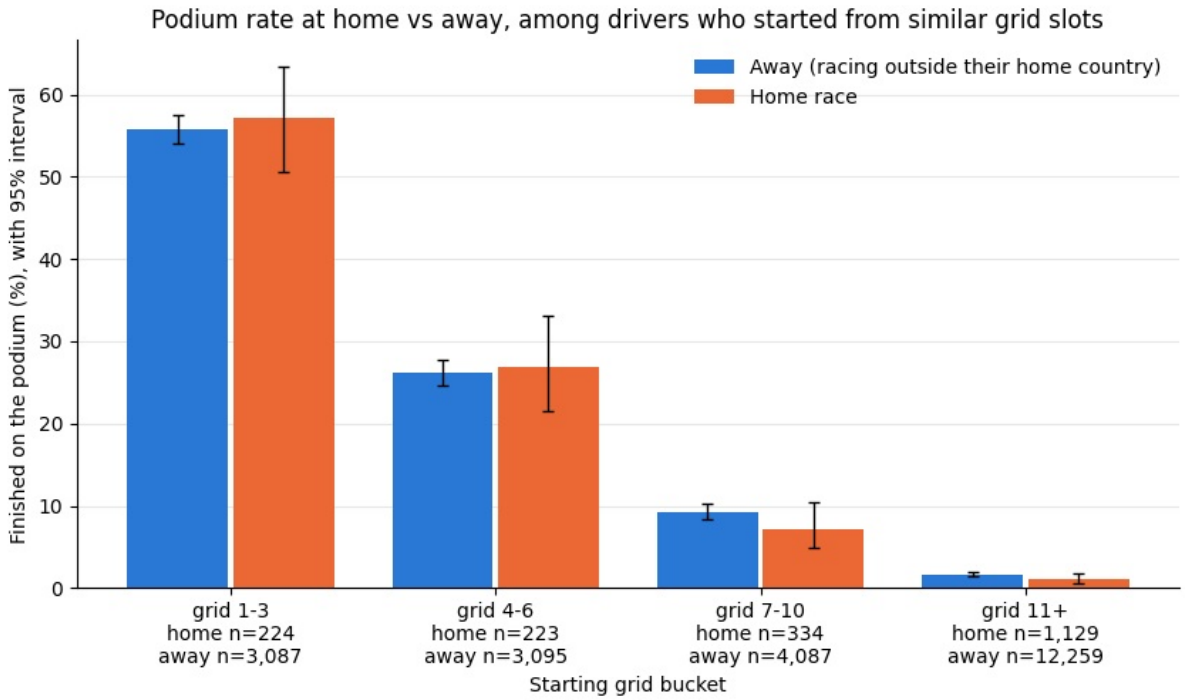



**Figure 5.** Q2: podium rate of home (orange) vs away (blue) drivers inside each grid bucket, 95% Wilson intervals; sample sizes in the axis labels.

### Q3. Which constructors have the highest mechanical-retirement rate in 2014–2021 vs 2022–2024?

**Status mapping (Decision 11):** 65 statuses among 2014–2024 starters, all mapped. Fixed by the TA: Puncture, Tyre, Tyre puncture, Fire, Safety belt = mechanical; Refuelling = not mechanical. Our rule for the rest: failure of a named car component (Engine, Gearbox, Power Unit, Brakes, Hydraulics …) = mechanical; collisions, accidents, spins, disqualifications

= not; 9 unclear statuses (Retired, Wheel, Damage, Debris, Out of fuel …, 77 starts) are not mechanical in the main answer and counted in a “broad” check. **Renamed teams (Decision 7)** are linked into lineages (e.g. Force India → Racing Point → Aston Martin) for Q3 only. Rate = mechanical retirements ÷ starts.

**Answer.** Overall the rate fell from **9.2%** to **5.5%** of starts. **2014–2021:** highest RB / Toro Rosso 14.7% (47 of 319), Alpine / Renault 13.5%, McLaren 13.0%; lowest Mercedes 4.1%. **2022–2024:** highest Alpine 10.4% (14 of 135), Sauber / Alfa Romeo 8.8%, RB 5.9%; lowest Aston Martin, McLaren and Red Bull (3.7% each). 8 of 10 teams improved; intervals separate the periods only for McLaren and (barely) RB. The answer is robust: strict vs broad mapping rank the teams almost identically (Spearman 0.91 / 0.95, same top 3). *Grain:* starts. *Cutoff:* status is post-race and is never a model feature.

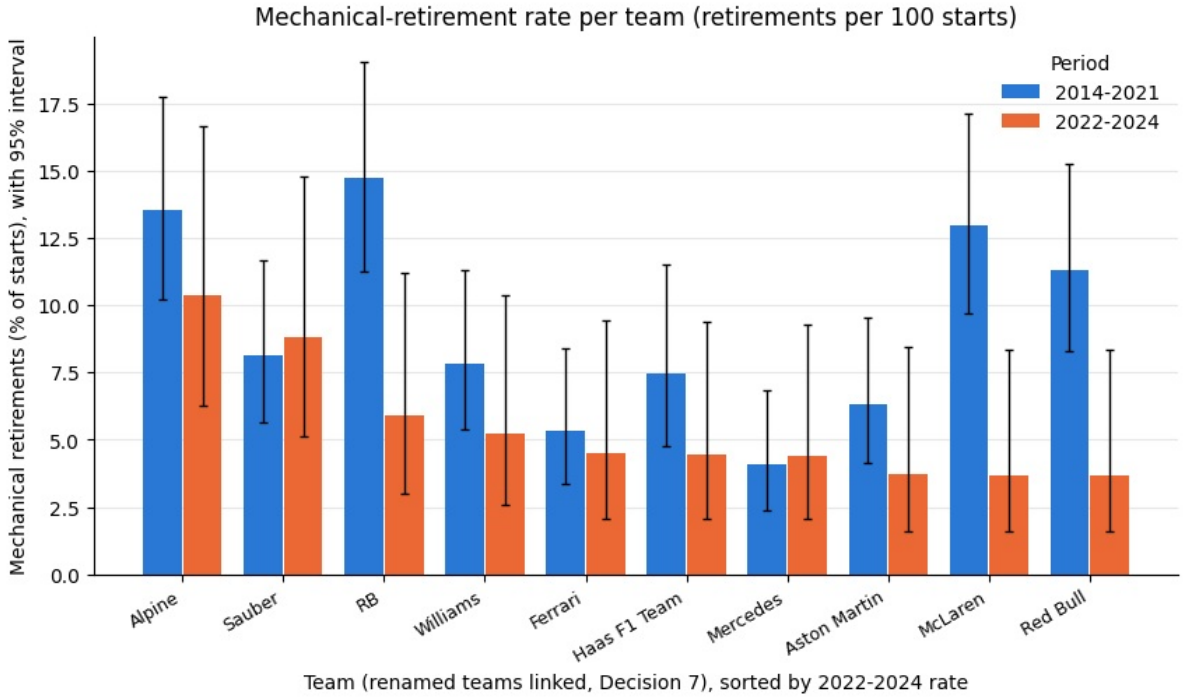



**Figure 6.** Q3: mechanical-retirement rate per team (lineage), 2014–2021 vs 2022–2024, 95% Wilson intervals.

## 4. Features and why each is valid at the pre-race cutoff

**Modelling table:** `model_df` = 8,910 rows (428 races, 2003–2024), podium rate 14.4%, one row per (raceId, driverId) (asserted). Form and standings are computed on **all** seasons so that 2003 already has history, but every feature only looks at **earlier** races. Missing values stay NaN and are filled inside the model Pipeline (training medians + “was missing” indicators). Feature-justification charts and trials use only seasons up to 2021 (train + validation) — the test seasons are never looked at.

| Group | Feature | What it is | Why it is known before the start |
|---|---|---|---|
| grid | grid_position | starting slot; pit lane = last slot + 1 (Decision 17) | the grid is fixed before the start |
| | grid_front_row, grid_pitlane_start | flags for slot 1–2 and pit-lane start | from the grid |
| quali | quali_position | qualifying position | qualifying ends before the race |
| | quali_q1_gap_pct | % slower than the fastest Q1 time of the race | qualifying ends before the race |
| | quali_grid_penalty | grid slot − qualifying position (> 0 = moved back) | both known before the start |
| drv | drv_podium_rate / avg_finish / dnf_rate _last10 | driver’s results in his previous 10 races | `.shift(1)`: earlier races only |
| con | con_podium_rate / avg_finish / dnf_rate _last10 | team’s results (both cars) in its previous 10 races | `.shift(1)`: earlier races only |
| stand | stand_driver / con _points_before, _rank_before | championship points and position before the race | standings of the **previous** round |
| ctx | ctx_home_race | circuit in the driver’s home country (Decision 12) | nationality and circuit are fixed |
| | ctx_field_size, ctx_driver_age, ctx_driver_races_before | starters, age on race day, earlier starts | known when the grid forms / fixed |

**Feature decisions.**

- **Decision 17 — pit-lane starters** (73 rows, 0 podiums): `grid_position` = last slot + 1 plus a flag. Raw 0 would sit in front of pole; NaN would be filled with a median (about 10th) — both wrong.
- **Decision 18 — form window:** last **10** races, across season boundaries, restarting after a break of more than a year (otherwise Mercedes’ first 2010 race would get its 1955 results as “form”). Trial on 2003–2021, ROC-AUC of the podium rate alone (driver / team): N = 3: 0.822 / 0.856, 5: 0.855 / 0.873, **10: 0.875 / 0.882**, 15: 0.882 / 0.886, 20: 0.881 / 0.883 — 10 is within 0.01 of the best and reacts within a season. No history → NaN (not 0, which would claim “never on the

podium”).

- **Decision 19 — constructor form follows `constructorId`:** the Q3 lineage map covers only 2014+ renames; linking would treat renames two ways, and both versions differ in only 1.1% of rows with the same AUC (0.8817 vs 0.8820).
- **Decision 20 — no year / era feature:** with a time split every test season is unseen, and the podium base rate is flat (12.6–15.8% per season).
- **Decision 9 — sprint races: use partly.** 18 sprint weekends (2021–2024). The sprint is finished before the Sunday start, so the Sunday grid (which the sprint set in 2021–22) is known at prediction time — not leakage. We keep the races and the grid but build no sprint features (they would be NaN in 96% of rows, all in validation/test). Standings from the previous round never include the current weekend’s sprint points.
- **Qualifying outlier:** one Q1 time of 1,002.6 s (1995) is not a real lap; a guard (Q1 above 1.5 × the fastest) sets such gaps to NaN, and it falls outside the 2003+ table anyway.

**Leakage checks (8.12):** driver and team form recomputed by brute force from races with an earlier date equal the features on 300 random rows; standings: “points before + race points = standings after” holds in 100% of normal-weekend rows (proving the table includes the race itself); asserts that no post-race column is a feature.

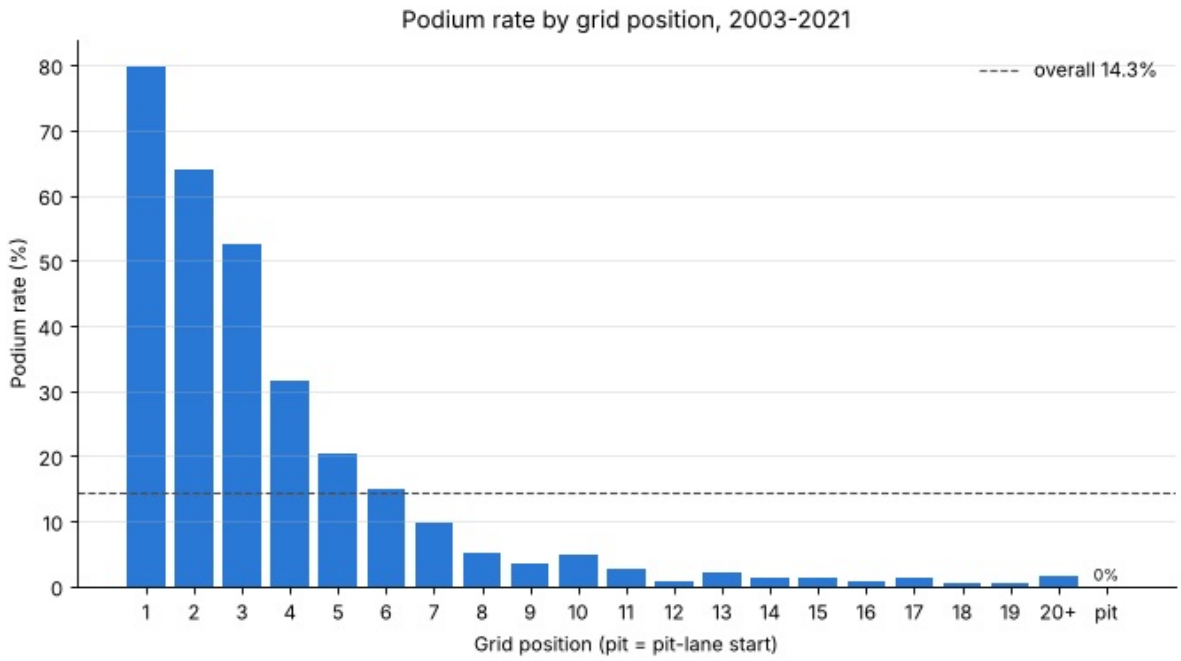



**Figure 7.** Podium rate by grid position, 2003–2021: 79.9% from pole, 5% or less from 8th; pit-lane starts 0%.

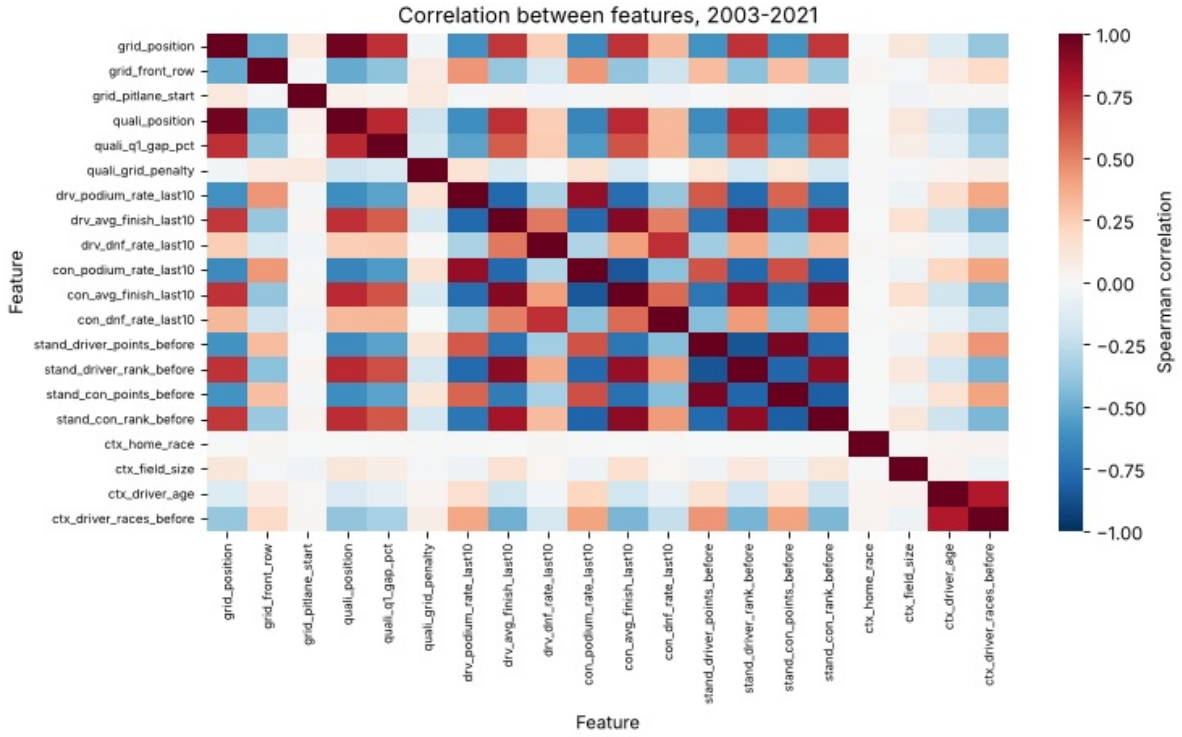



**Figure 8.** Spearman correlation between the 20 features (2003–2021): strong overlap within and between groups (grid vs qualifying position 0.97).

The other group charts (notebook 8.13b–f) show the same monotone patterns: within 0.25% of the fastest Q1 time → 55.2% podium; driver with 9–10 podiums in his last 10 races → 71.2% vs 2.2% with none; team podium share above 0.5 per car → 59.7%; championship leader before the race → 69.0%; experience peaks at 151–200 earlier starts (29.9%).

## 5. Train / validation / test split (Decision 10)

We predict future races from past ones, so the split is **by season**. A random 70/15/15 row split would put drivers of **100%** of the test races into training too, and 99.8% of training rows would come from races later than the first test race. Three boundary options were compared (the brief's "≤ 2019 / 2020–21 / ≥ 2022" is only a naive example, per the TA):

| Option | Train | Validation | Test | Val rows | Test rows |
|---|---|---|---|---|---|
| A | 2003–2019 | 2020–2021 | 2022–2024 | 775 | 1,352 |
| **B (chosen)** | **2003–2018** | **2019–2021** | **2022–2024** | **1,195** | **1,352** |
| C | 2003–2019 | 2020–2022 | 2023–2024 | 1,214 | 913 |

Podium share is 14.1–15.1% in every split of every option, so balance does not decide. Option B gives a validation set of three seasons in which the short, unusual COVID season 2020 (17 races at 14 circuits) is 28% of the rows instead of 44% (option A), and keeps the **whole 2022 ground-effect rules era as an untouched test set** — an honest "forecast after a rule change". Final sizes: train 6,363 rows (300 races, 14.1% podium), validation 1,195 (60 races, 15.1%), test 1,352 (68 races, 15.1%).

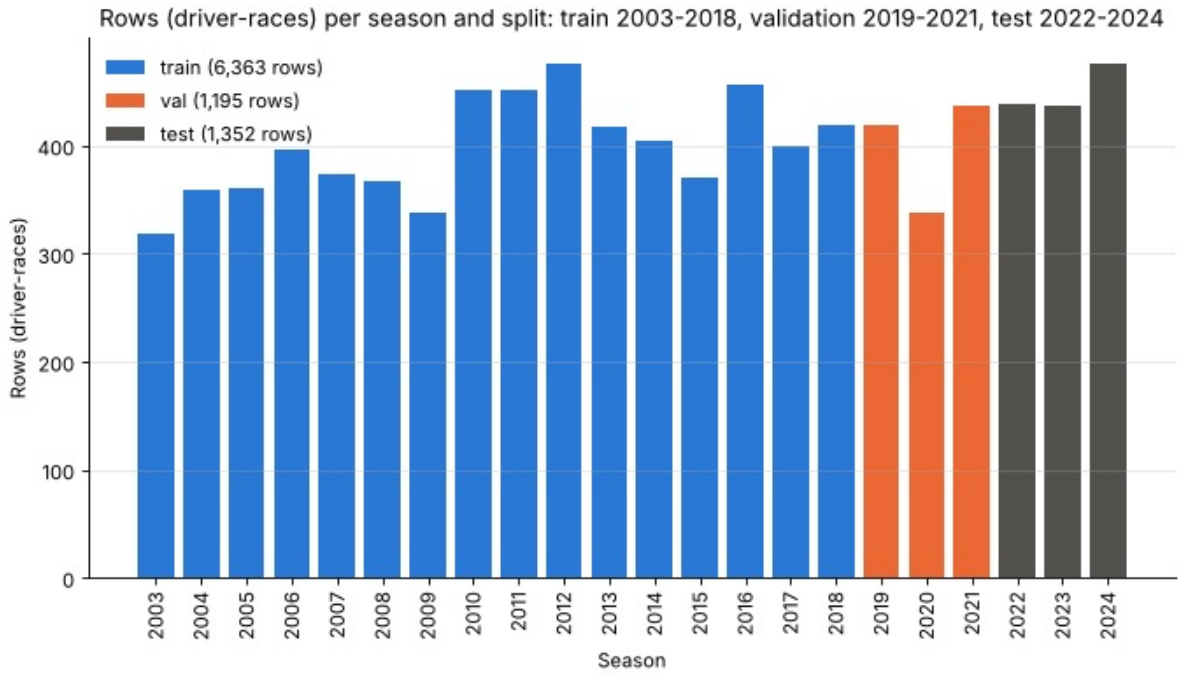



**Figure 9.** Rows per season and split: train 2003–2018 (blue), validation 2019–2021 (orange), test 2022–2024 (grey).

## 6. Models and results

Every model is a scikit-learn `Pipeline`: median imputer with “was missing” indicators (fitted on train only), plus a scaler for logistic regression and the network (Figure below shows the effect: 1,400 missing training cells → 0, 20 → 31 columns, spreads differing about 3,000× → all standard deviation 1). XGBoost handles missing values itself. For every model the probability threshold is the one with the best F1 **on validation**, applied unchanged to test.

- **Rule to beat:** “podium if grid ≤ 3” (score = 1 / grid position).
- **Logistic regression (baseline, per the TA):** weighted sum of scaled features → sigmoid probability; regularised by C.
- **Random forest:** 300 decision trees on random samples of rows and features, probabilities averaged.
- **XGBoost:** gradient boosting — small trees built one after another, each correcting the remaining errors.
- **Shallow neural network (Keras):** 1–2 hidden ReLU layers + sigmoid output, Adam optimiser, dropout and early stopping on the last 15% of training rows (validation seasons stay unused).

Each family has three labelled attempts (default → tuned on validation → class weights / dropout), 13 attempts in total.

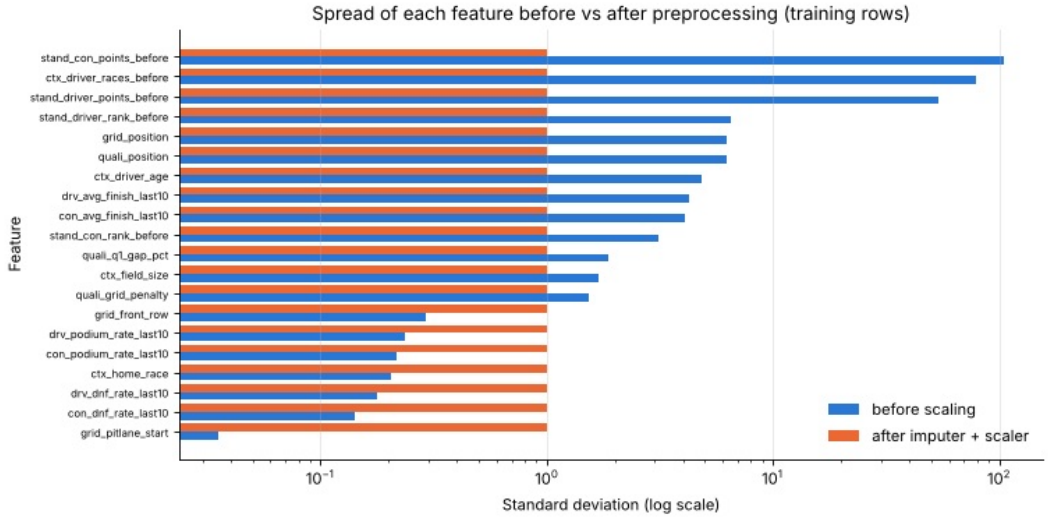



**Figure 10.** Effect of the preprocessing on the training rows: standard deviation of each feature before (blue) and after imputation and scaling (orange), log scale.

| Model (best attempt of each family) | Train PR-AUC | Val PR-AUC | Val ROC-AUC | Val F1 | Test PR-AUC | Test ROC-AUC | Test F1 | Test precision | Test recall |
|---|---|---|---|---|---|---|---|---|---|
| Rule: grid ≤ 3 | 0.620 | 0.683 | 0.917 | 0.719 | 0.647 | 0.913 | 0.649 | 0.650 | 0.647 |
| Logistic regression (LR_baseline) | 0.690 | 0.726 | 0.940 | 0.728 | 0.719 | 0.937 | 0.679 | 0.650 | 0.711 |
| **Random forest (RF_v2_tuned) — final** | **0.766** | **0.748** | **0.938** | **0.735** | **0.719** | **0.939** | **0.682** | **0.652** | **0.716** |
| XGBoost (XGB_v2_tuned) | 0.729 | 0.734 | 0.941 | 0.748 | 0.701 | 0.937 | 0.670 | 0.615 | 0.735 |
| Neural network (NN_v3_tuned) | — | 0.715 | 0.936 | 0.722 | 0.698 | 0.930 | 0.653 | 0.562 | 0.779 |

All 13 attempts with all five metrics on train, validation and test are in notebook cell 10.11. Notable attempts: the default forest overfits (train PR-AUC 1.000, validation 0.719); the default XGBoost too (train 0.894, validation 0.706) until the trees are limited to depth 2 (0.734); class weights never improve PR-AUC (LR 0.718, RF 0.742, XGB 0.732); dropout helps the network (0.692 → 0.708 → 0.715 tuned).

**Decision 21 — final model and threshold:** `RF_v2_tuned` (300 trees, at least 20 rows per leaf, √features per split), the attempt with the highest validation PR-AUC; threshold 0.374 (best validation F1: precision 0.702, recall 0.772). “Top 3 probabilities per race” was compared on validation (F1 0.717) but not used, because the inference function scores one driver at a time.

**Test result.** On the 2022–2024 seasons the final model reaches PR-AUC **0.719** (validation 0.748), ROC-AUC 0.939, F1 0.682: 146 of 204 podiums found with 78 false alarms. The drop from validation is expected — the test seasons are a new rules era — and every learned model still beats the grid rule by about 0.07 PR-AUC.

**Limitations of each model.** Logistic regression is linear (cannot follow patterns like experience rising then falling) and its coefficients are hard to read with correlated features; forests and boosted trees overfit unless leaves/depth are limited and their probabilities are not perfectly calibrated; the network has little data (about 5,400 rows after the early-stopping split) and depends on the seed (fixed).

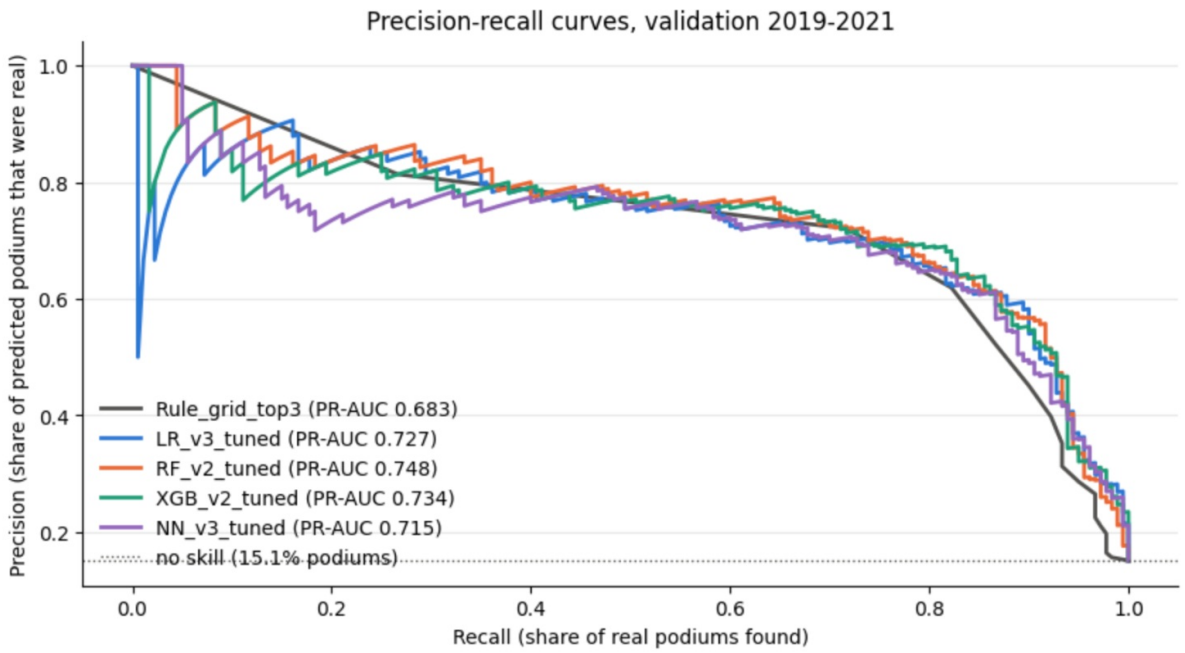



**Figure 11.** Precision-recall curves on validation (2019–2021), best attempt of each family; dotted line = no skill (15.1% podiums).

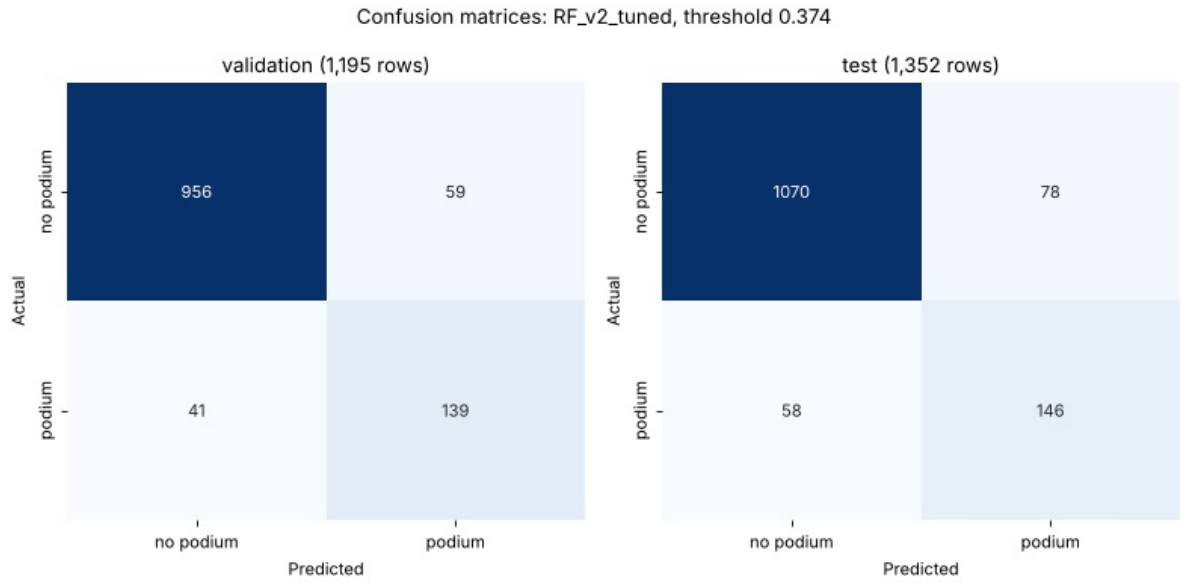



**Figure 12.** Confusion matrices of the final model at threshold 0.374, validation (left) and test (right).

ROC-AUC is above 0.9 for every model including the rule (notebook 10.10), because the back of the grid is easy to rule out; PR-AUC shows the harder part — picking the three podium drivers among the front runners — which is why it is the main metric.

## 7. Experiments

All experiments use the best scikit-learn model (`RF_v2_tuned`, validation PR-AUC 0.748) and are scored on validation only.

**Feature-group ablation.** Removing the **grid** group costs the most (0.748 → 0.716, −0.033): the grid after penalties is the information closest to the start. Driver form (−0.009), constructor form (−0.008), qualifying (−0.005) and standings (−0.005) cost little alone because they overlap (they all measure how strong car and driver are). Without the context group the model is slightly better (+0.004) — within noise for 180 validation podiums, consistent with Q2 (no home effect).

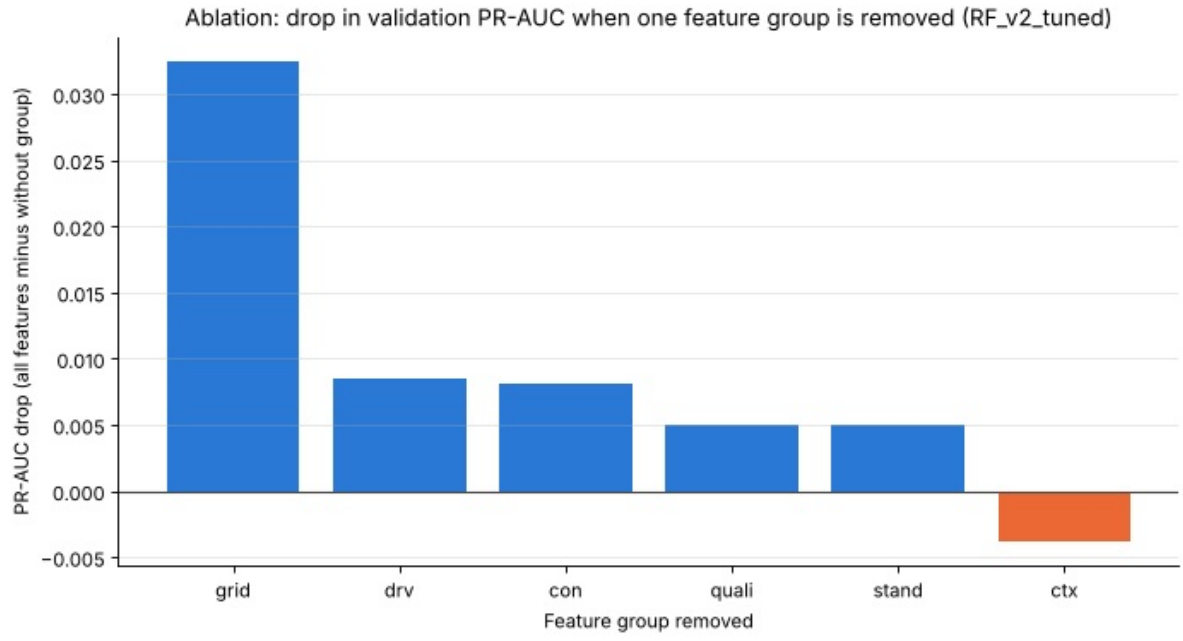



**Figure 13.** Ablation: drop in validation PR-AUC when one feature group is removed (orange = slightly better without it).

| Experiment | Setting | Validation PR-AUC | What it shows |
|---|---|---|---|
| 1. Imputer + scaler | fitted on train only (our Pipeline) | 0.7262 | clean estimate |
| | fitted on all seasons before the split | 0.7264 | 9 of 20 medians change (e.g. team points 21 → 26): the future shapes the transformation; tiny here, larger after rule changes |
| 2. Oversampling podiums to 50/50 | none | 0.748 | reference |
| | after the split (train only) | 0.739 | no gain — changes probabilities more than ranking |
| | before the split | **0.935** | fake: “validation” becomes 52% podiums with copied rows |
| Train both ways (Decision 4) | train 1994–2018 (18.6% without qualifying) | RF 0.738 / LR 0.720 | more rows, worse |
| | train 2003–2018 (0.3% without qualifying) | **RF 0.748 / LR 0.726** | start in 2003 is confirmed |

## 8. Explainability

> **Important:** “important to the model” does **not** mean “causes podiums”. These methods show which inputs the model *relies on* to make its predictions. The features are correlated with each other and with things the data does not contain (car speed, budget, team quality); a good grid slot is a *result* of a fast car and driver, not a cause we could change.

**Global — permutation importance (final model, validation):** shuffling `grid_position` costs 0.062 PR-AUC, `quali_position` 0.036, `grid_front_row` 0.021; form, standings and the Q1 gap about 0.01 each; pit-lane, home race and field size about 0.

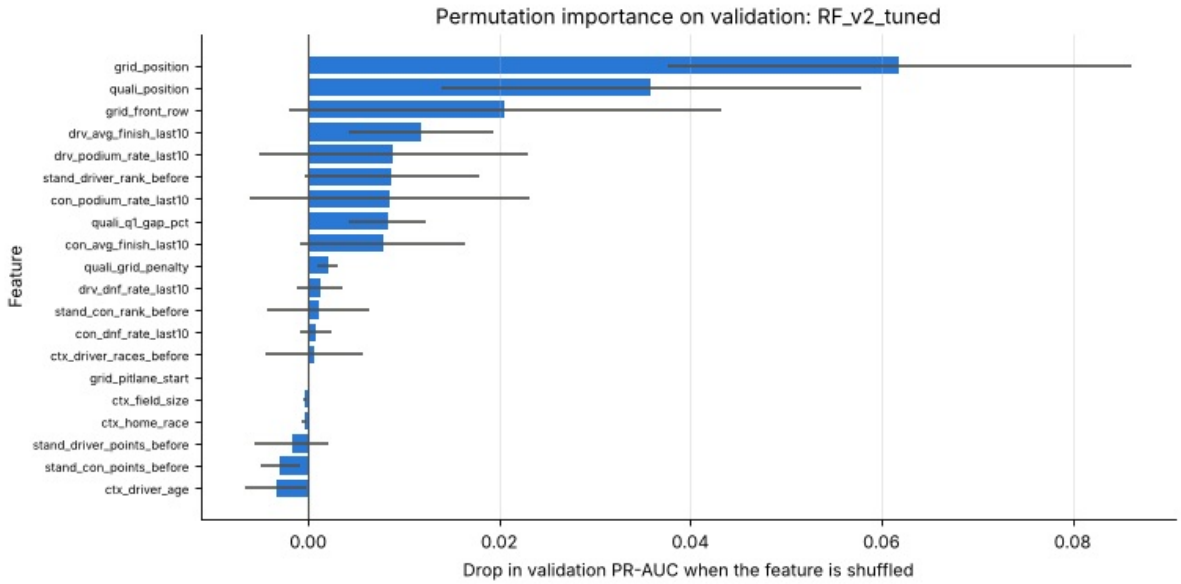



**Figure 14.** Permutation importance of the final model on validation (mean drop in PR-AUC over 10 shuffles, ± 1 standard deviation).

**Global — SHAP** (exact TreeSHAP of the tuned XGBoost on 500 validation rows): the inputs that move predictions most are qualifying position (mean |SHAP| about 0.73 log-odds) and grid position (0.55), then team and driver average finish and recent podium rates. Low (front) positions push towards a podium, high ones push away. The same groups lead as in the permutation importance; grid and qualifying position swap places because they are 0.97 correlated and each model leans on one of them a bit more.

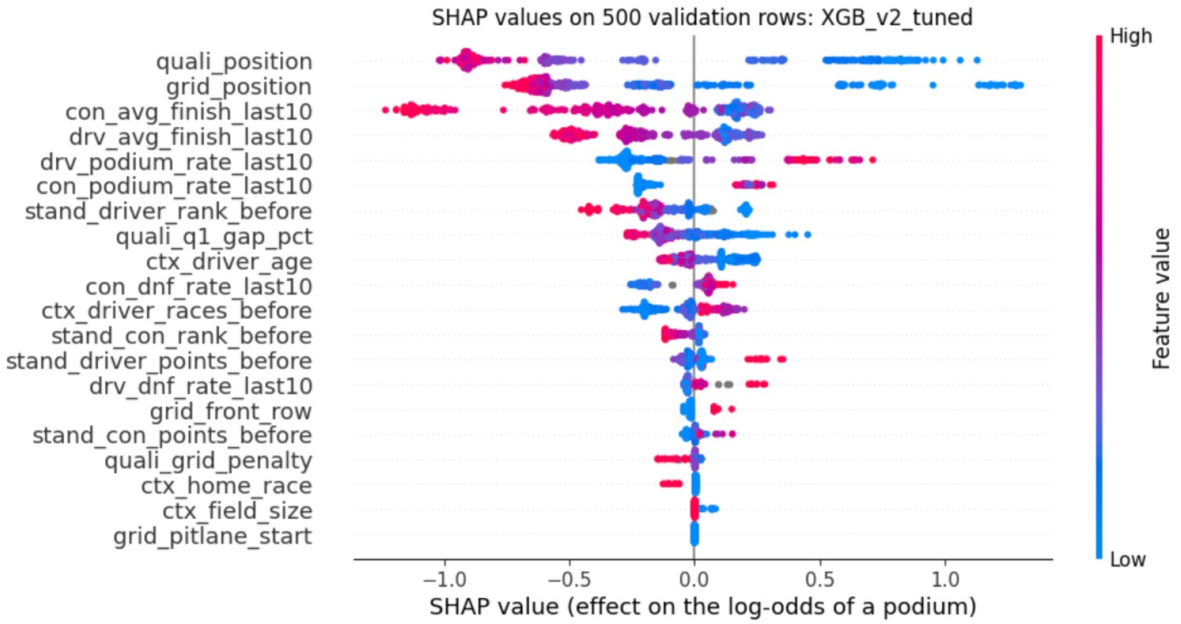



**Figure 15.** SHAP beeswarm (XGB_v2_tuned, 500 validation rows): one dot per row; right = towards a podium; colour = feature value (red high, blue low).

**Local — LIME** (final model) on two validation cases:

- **Surprise podium: Daniil Kvyat, 2019 German GP** — started 14th, model probability 0.002, finished 3rd in a chaotic wet race. LIME: not on the front row (−0.093), grid 12–17 (−0.039), no team and no driver podium in the last 10 races (−0.032 each), qualified 12th–16th (−0.031). The model gave him a low chance for the right pre-race reasons; the race events that produced the podium are, by design, not features.
- **Missed favourite: Lewis Hamilton, 2020 Italian GP** — pole, probability 0.838, finished 7th after a stop-go penalty. LIME: front row (+0.097), grid ≤ 6 (+0.093), qualified ≤ 6th (+0.086), strong driver and team form (+0.055, +0.049).

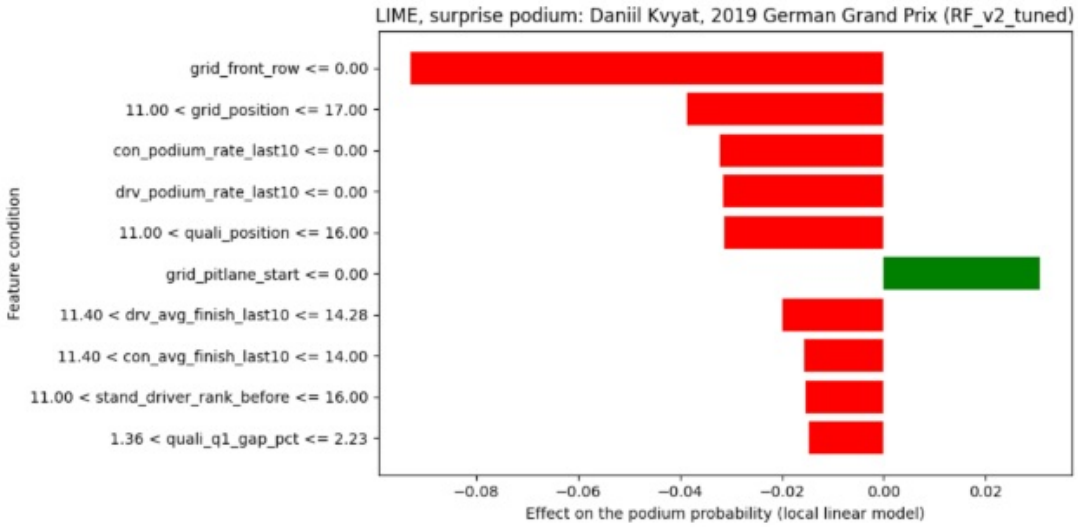



**Figure 16.** LIME explanation of the final model for the surprise podium (Kvyat, 2019 German GP): red pushes the probability down, green up.

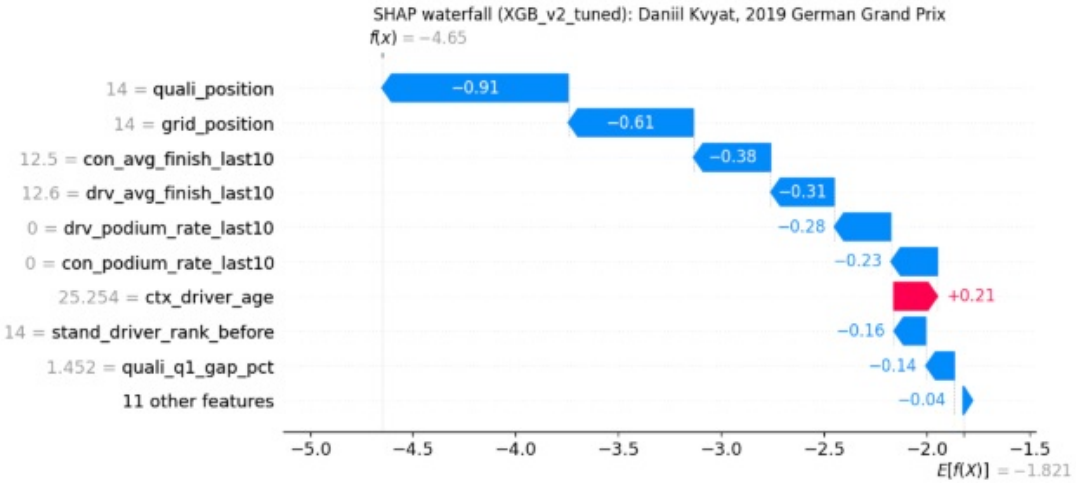



**Figure 17.** Local SHAP waterfall (XGBoost) for the same driver: from the average log-odds −1.82 to −4.65; qualifying 14th (−0.91) and grid 14th (−0.61) push down most.

## 9. Inference function

`predict_podium(record)` takes one driver-race as a dictionary of feature values, builds a one-row table with exactly the 20 features (anything missing or `None` becomes NaN), and returns the probability from the saved final Pipeline (training-fitted preprocessing) and a yes/no answer at the validation threshold 0.374. It refuses unknown or post-race columns (e.g. `positionOrder`).

| Demo | Input | Output |
|---|---|---|
| Complete record | Max Verstappen, 2021 Abu Dhabi GP (pole, all 20 features) | probability 0.835 → podium; equals the Pipeline's own prediction (assert) |
| Partially missing record | rookie, grid 6, no qualifying time, no driver history (7 features missing) | probability 0.094 → no podium; missing features listed |

## 10. Known limitations

- **Qualifying-coverage drift:** qualifying is complete only from 2003, so the model covers 2003–2024 only; earlier eras (different reliability and field sizes) are out of scope.
- **Era sensitivity:** the test seasons are the 2022 rules era; scores drop from validation (0.748) to test (0.719). Retirement rates and car performance shift between eras.
- **Small validation set:** 180 podiums — differences below about 0.01 PR-AUC between models or ablation settings are within noise.
- **Race events are invisible by design:** crashes, safety cars, penalties and weather (e.g. Kvyat 2019) cannot be predicted before the start.
- **Judgement calls:** status → mechanical mapping (partly TA-fixed, robust in a broad check), nationality dictionary, team-lineage map (Q3 only), duplicate rule, Indy 500 exclusion, pit-lane encoding, form window.
- **Sprint weekends 2021–22:** the qualifying table describes the sprint grid, not the Sunday grid.
- **Data quirks:** grid and qualifying almost never differ before 2004 (penalties probably not recorded); fastest-lap columns exist only from 2004; no engine-supplier or tyre data.
- **Small samples in Q1–Q3:** 10–26 front-row starts per circuit after 2014; about 135 starts per team in 2022–2024.

## Appendix: decision log

| # | Decision | Evidence |
|---|---|---|
| 3 | PR-AUC main metric, ROC-AUC secondary; F1, precision, recall reported | 13.6% podiums (cell 4.9) |
| 4 | Modelling table 2003–2024 | qualifying coverage (4.1); train-both-ways 0.748 vs 0.738 (11.5) |
| 5 | Drop all rows of repeated driver-race pairs | grid can’t separate cars; first row follows finishing order |
| 6 | Exclude the Indianapolis 500 | 0.6% driver overlap, 99.8% American |
| 7 | Link renamed teams, Q3 only | 6 vs 10 comparable teams; renames within noise |
| 8 | Drop non-starters (F/W), keep 87 pit-lane starters | entries vs starters (4.4), grid = 0 (4.6, 4.7) |
| 9 | Sprints: use partly (Sunday grid, no sprint features) | grid = qualifying in 12–18% of 2021–22 sprint rows |
| 10 | Split 2003–18 / 2019–21 / 2022–24 | random split leaks (9.1); option table (9.2); 2020 check (9.3) |
| 11 | Status → mechanical mapping | TA ruling + component rule; strict vs broad Spearman 0.91/0.95 |
| 12 | Nationality → country dictionary | 43 nationalities asserted |
| 13–14 | Read CSVs as text; convert types in place; save clean.pkl | 160,144 markers → NaN; conversions checked |
| 15–16 | Q1 minimum 10 starts; Q2 grid buckets + race bootstrap | circuits kept per minimum; bucket rates |
| 17 | Pit lane = last slot + 1 + flag | 0 podiums from 87 pit-lane starts |
| 18 | Form over last 10 races, restart after 1-year break, NaN if no history | window trial (8.5) |
| 19 | Constructor form by constructorId | 1.1% of rows differ, same AUC (8.7) |
| 20 | No year/era feature | flat base rate; unseen test years |
| 21 | Final model RF_v2_tuned, threshold 0.374 | best validation PR-AUC (10.8) |